

Global Population Dynamics and Forecasting

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Summary

This project analyzes global population trends using data from the World Population dataset and the World Bank Group. The goal is to understand how populations are changing over time and what patterns emerge across countries and regions.

The analysis focuses on three main areas:

- Finding which countries have gained or lost the most population, both historically and in the past five years.
- Examining annual population growth rates to see which countries grow the fastest or slowest, and how much these rates vary.
- Comparing population growth across world regions and income groups to see how geography and economic conditions relate to demographic change.

About the Data

This project uses publicly available datasets from the World Bank Group (<https://data.worldbank.org/>) to analyze global population trends.

The following indicators were collected:

- Total Population: <https://data.worldbank.org/indicator/SP.POP.TOTL>
- Death rate: <https://data.worldbank.org/indicator/SP.DYN.CDRT.IN>
- Birth rate: <https://data.worldbank.org/indicator/SP.DYN.CBRT.IN>
- Population distribution: <https://data.worldbank.org/indicator/SP.POP.1564.TO.ZS>
- Net Migration Data: <https://data.worldbank.org/indicator/SM.POP.NETM>

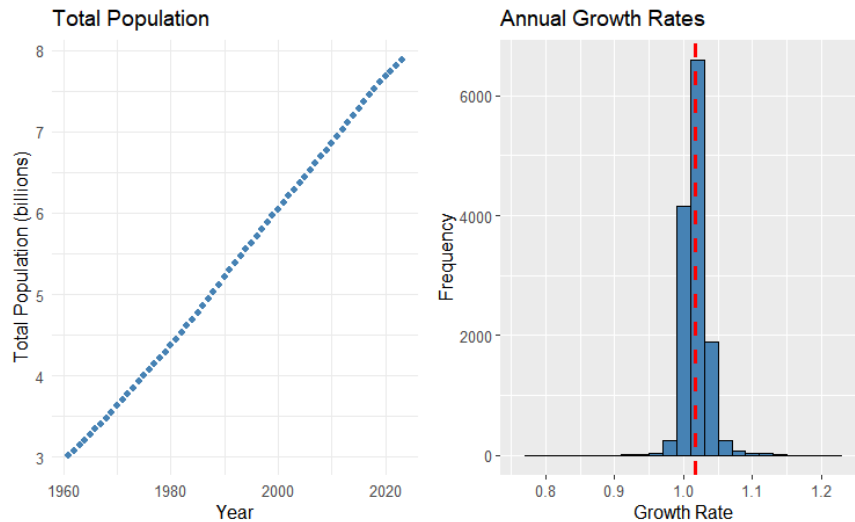
Data Cleaning and Preparation

The datasets were cleaned and reshaped into a long format to make them easier to analyze and visualize. Column names were standardized for consistency, and rows containing missing values were removed to maintain data quality. Country names were also validated using recognized country codes, ensuring that only valid entries remained in the final dataset.

Code	Name	Year	Population	BirthRate	DeathRate	Net_Migration	Percent_15_64	Region	IncomeGroup
AFG	Afghanistan	1,960	9,035,043	50.516	31.672	2,606	55.22988	Middle East & North Africa	Low income
AFG	Afghanistan	1,961	9,214,083	50.566	31.129	6,109	55.16205	Middle East & North Africa	Low income
AFG	Afghanistan	1,962	9,404,406	50.666	30.630	7,016	55.09435	Middle East & North Africa	Low income

Part 1: EDA of population growth

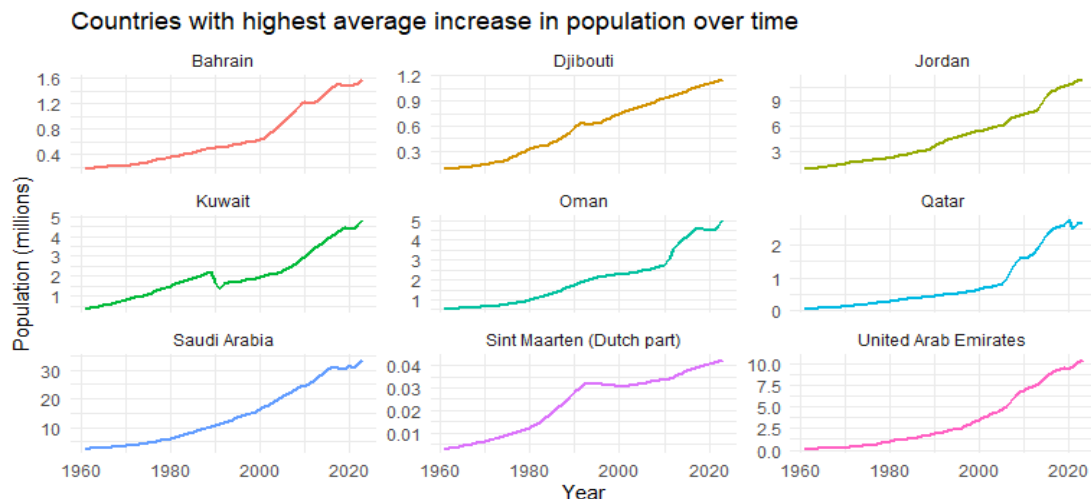
The annual population growth rate shows how the population in a given year compares to the year before. If this growth rate stays the same over several years, we can estimate the population in any future year by starting with the initial population and applying that same rate repeatedly. This idea is known as **geometric population growth**, where the population increases in fixed time steps at a constant rate.

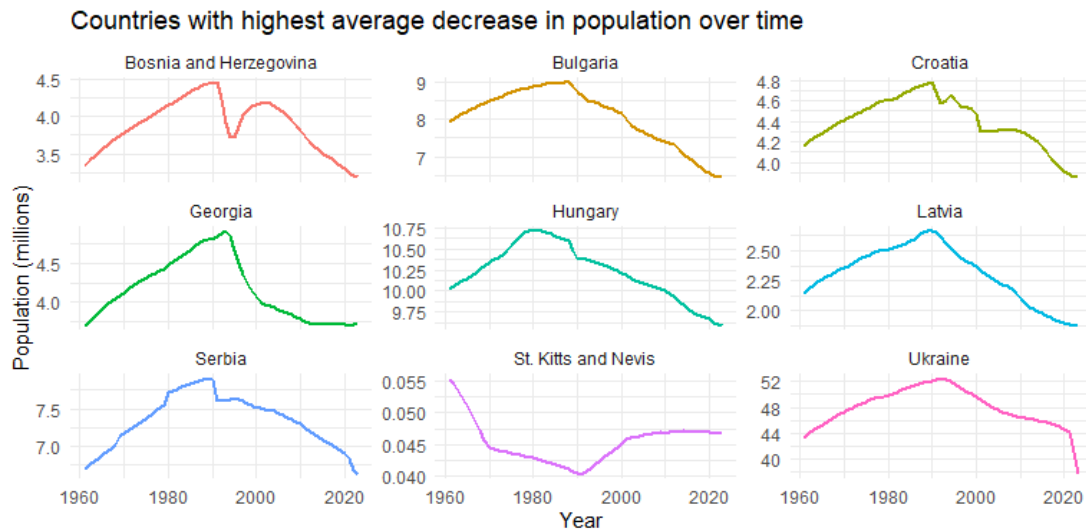


The Total Population plot shows that the world population grew steadily from 1960 to 2020, following a nearly linear upward trend. This suggests a fairly consistent rate of increase over time.

The histogram of annual growth rates shows that most countries have growth rates slightly above 1, meaning populations generally increased each year. A small number of countries experienced unusually fast growth, while a few had growth rates below 1, indicating population decline.

Which countries have experienced the highest increase/decrease in population over time?





Countries with the highest average increases in population all show clear upward trends from 1960 to 2020, with places like Saudi Arabia, the United Arab Emirates, Kuwait, and Qatar experiencing especially rapid growth. In contrast, countries with the highest average decreases display the opposite pattern: many grow until the late twentieth century and then decline steadily. Bulgaria, Croatia, Latvia, Serbia, and Ukraine show some of the most pronounced drops in population over time.

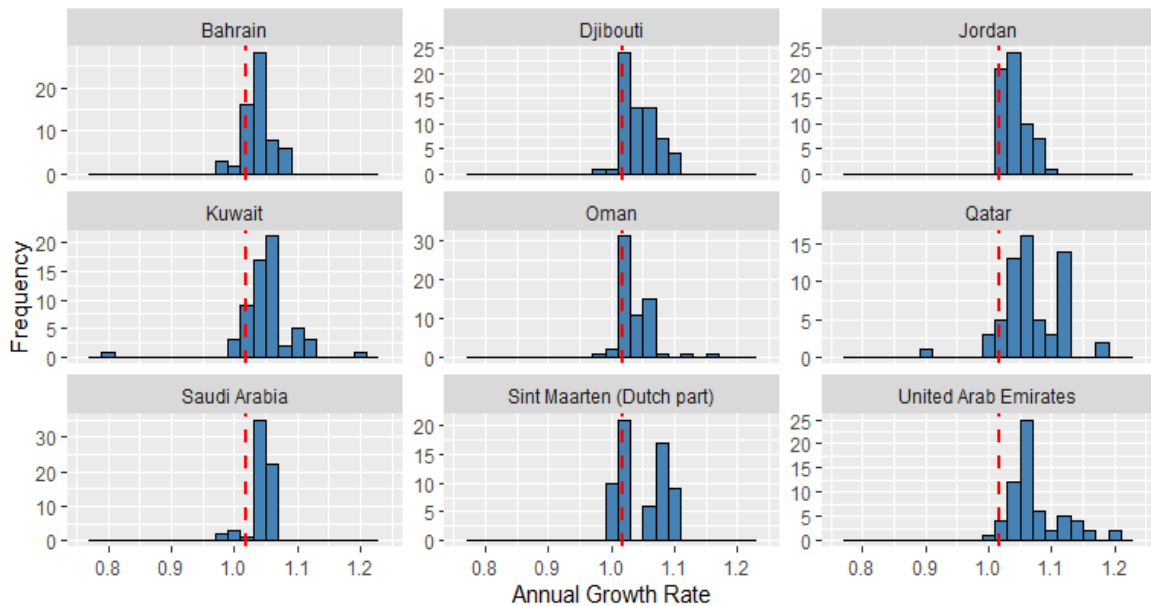
Which countries have been experiencing the highest increase/decrease in population in the last five years?

<i>Top Countries highest increase in population in last 5 years</i>			
Name	Pop_2020	Pop_2024	Growth_Percent
Seychelles	98,462	119,773	21.64388
Chad	17,224,679	19,319,064	12.15921
Syrian Arab Republic	21,049,429	23,594,623	12.09151
Oman	4,522,497	5,049,269	11.64781
United Arab Emirates	9,401,038	10,483,751	11.51695
Kuwait	4,400,267	4,853,420	10.29831
Niger	23,717,613	26,159,867	10.29722
Somalia, Fed. Rep.	16,651,191	18,358,615	10.25407
Congo, Dem. Rep.	95,989,998	105,789,731	10.20912

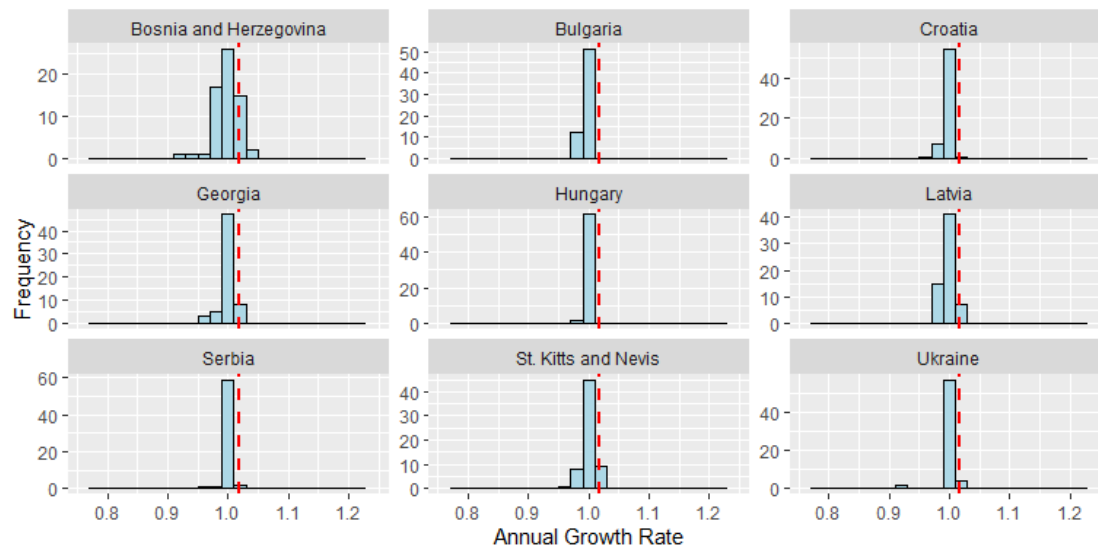
Countries with highest decrease in population in last 5 years

Name	Pop_2020	Pop_2024	Growth_Percent
Ukraine	44,680,014	37,732,836	-15.548737
St. Martin (French part)	31,786	27,515	-13.436733
Marshall Islands	42,706	38,827	-9.083033
Moldova	2,635,130	2,457,783	-6.730104
Tuvalu	10,399	9,816	-5.606308
Northern Mariana Islands	47,528	45,143	-5.018095
Qatar	2,794,148	2,656,032	-4.943045
American Samoa	49,761	47,521	-4.501517
Serbia	6,899,126	6,623,183	-3.999681

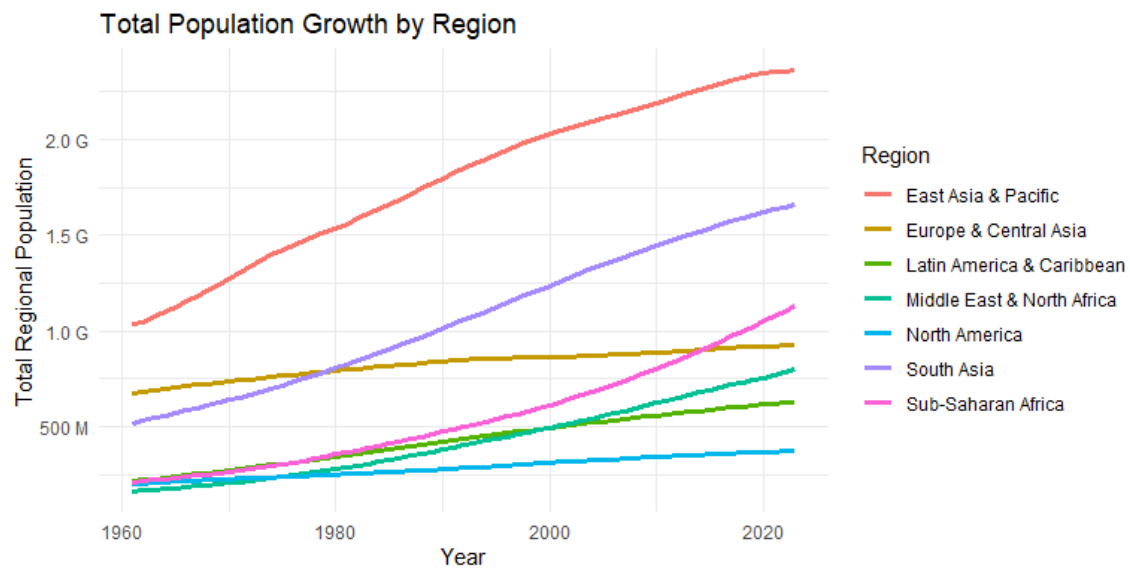
Histograms of annual growth rates with highest average rates.

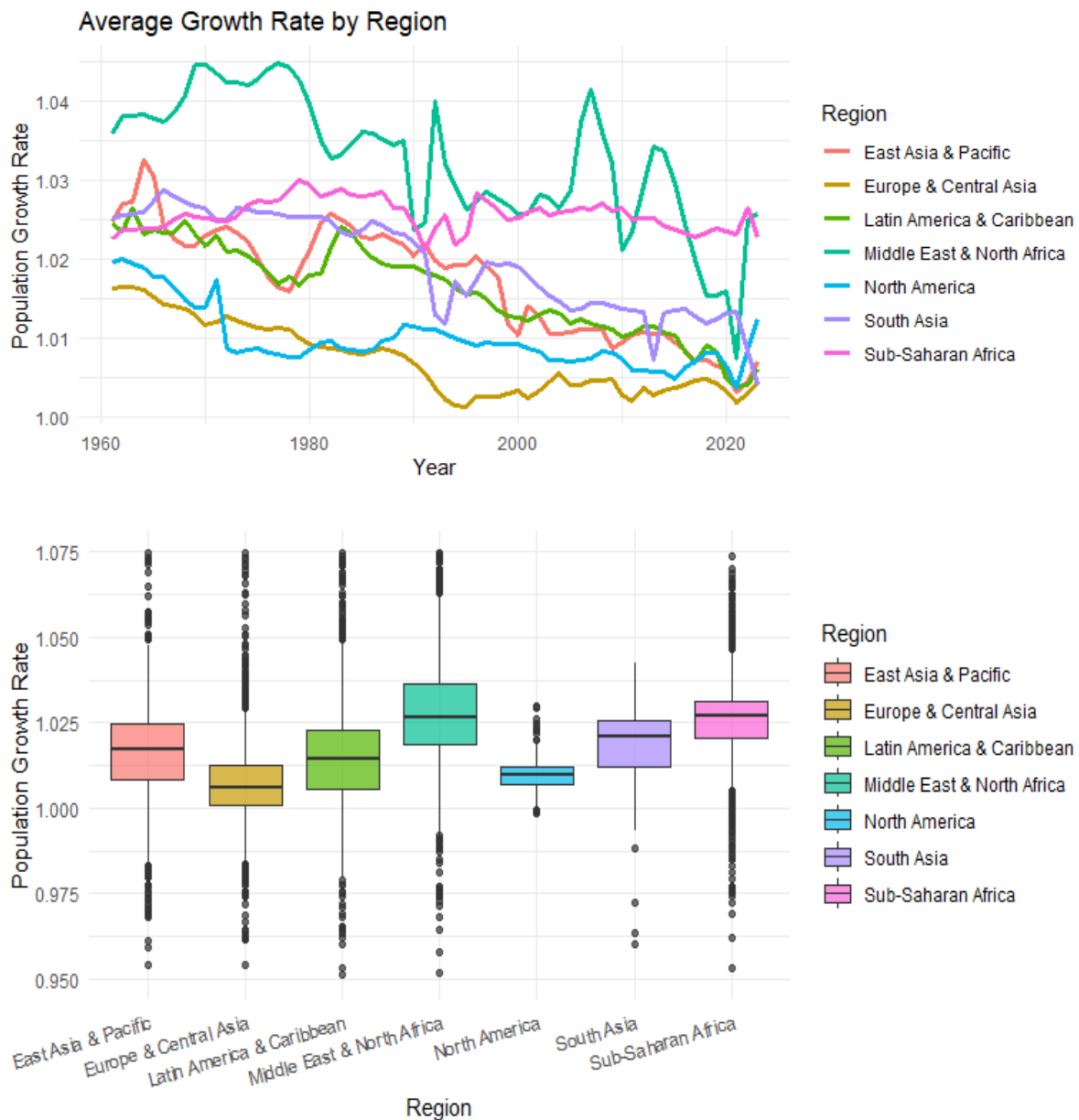


Histograms of annual growth rates with lowest average rates.



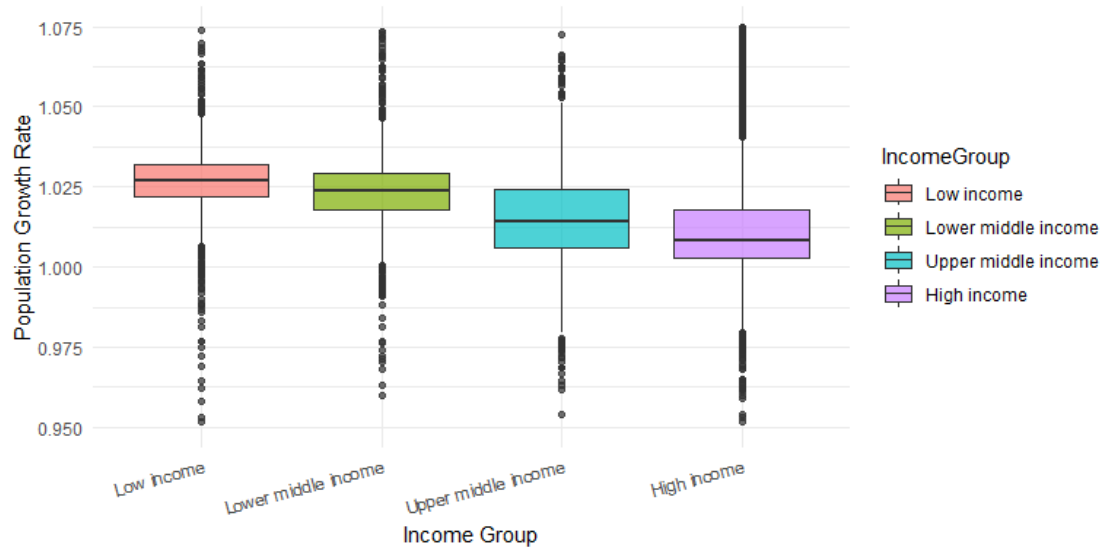
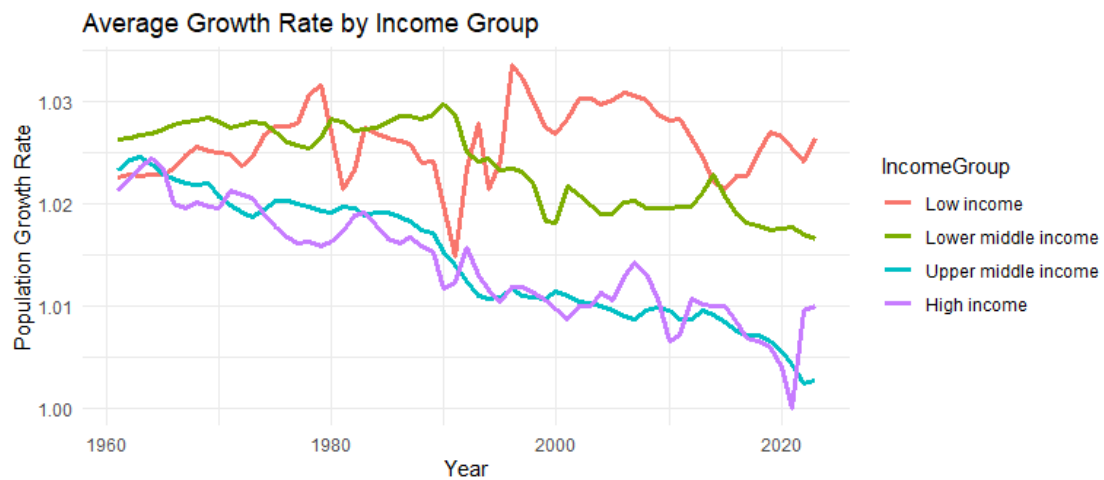
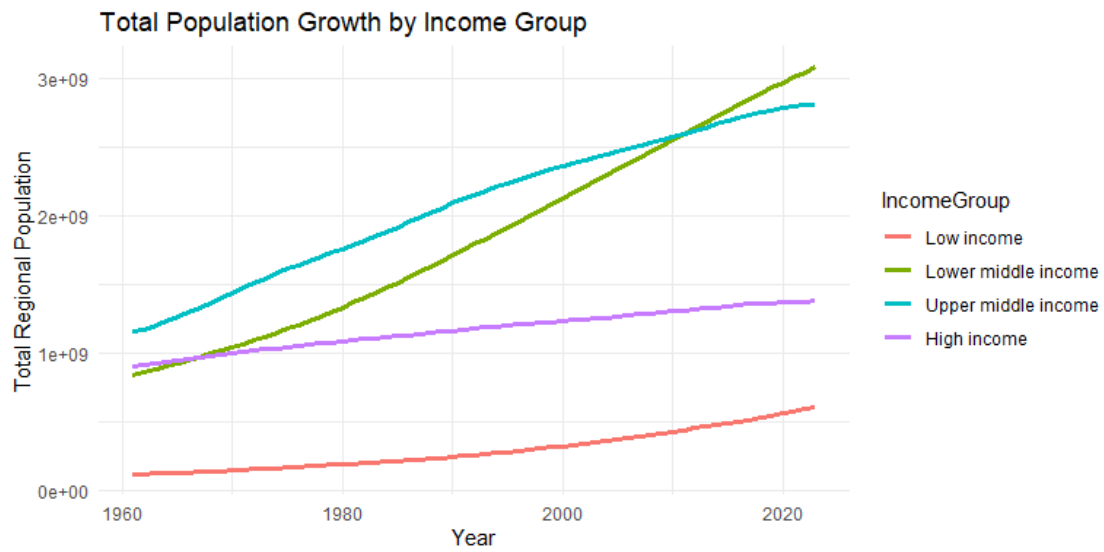
Population growth of countries over time grouped by region.





Across all regions, total population has increased steadily since 1960, with the largest rises seen in East Asia & Pacific, South Asia, and Sub-Saharan Africa. Although populations grew, average growth rates have gradually declined over time in most regions. Sub-Saharan Africa and the Middle East consistently show the highest growth rates, while Europe & Central Asia has the lowest. The boxplot comparison reinforces these patterns, showing clear regional differences in both the level and spread of population growth.

Population growth of countries over time grouped by income group.



Total population has increased across all income groups since 1960, with lower-middle-income and upper-middle-income countries showing the strongest overall growth. Despite rising populations, growth rates for all groups have gradually declined over time. Low-income and lower-middle-income countries maintain the highest growth rates, while high-income and upper-middle-income countries grow more slowly. The boxplots reinforce these differences, showing that growth is generally faster and more variable in lower-income groups compared to higher-income ones.

Part 2: Exponential Population Growth Model

In an exponential growth model, the population increases continuously at a constant rate. This rate, called the instantaneous growth rate, reflects how fast the population would grow at any moment based on the balance between births and deaths.

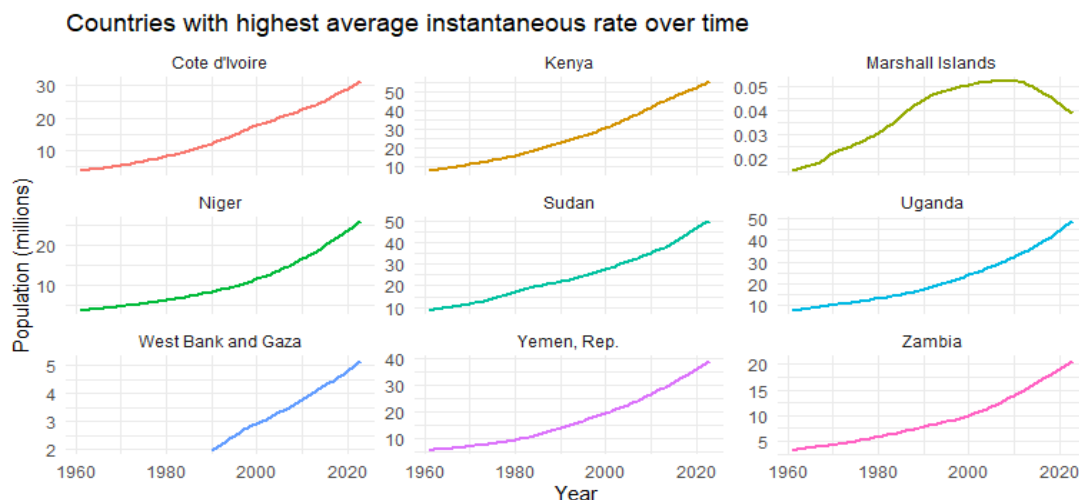
Exponential and geometric growth models behave similarly. The difference is that the exponential model treats growth as continuous, while the geometric model treats it as happening in discrete steps from year to year. However, the two models can be converted back and forth by using logarithms or exponentials.

The instantaneous growth rate for a specific year describes how much the population changes from one year to the next. When we look at multiple years, the overall growth rate is essentially the average of these yearly rates. In practice, this means that the long-term growth rate is determined by how the final population compares to the initial population, spread evenly across the number of years involved.

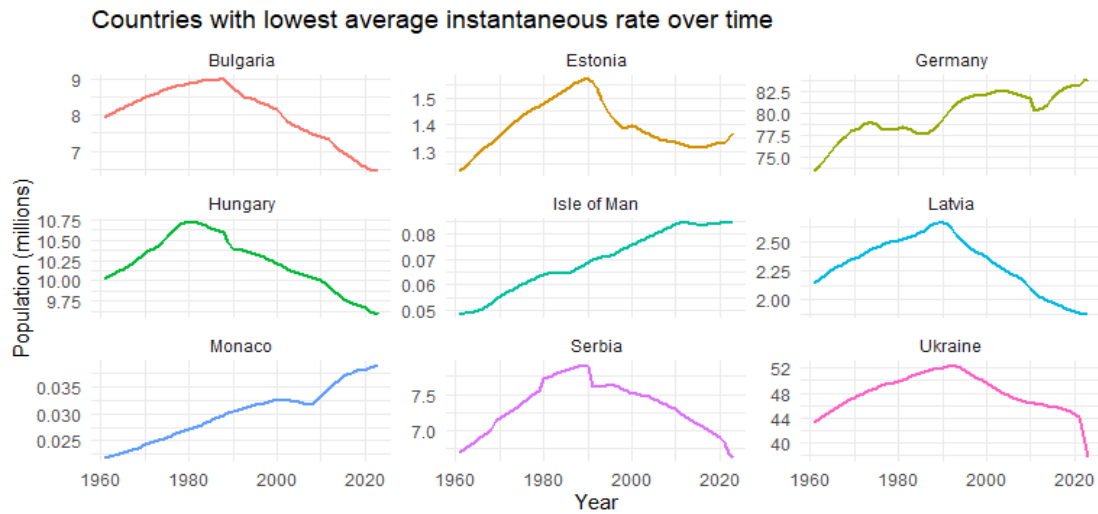
Similarly, the average annual growth factor over many years is the geometric mean of the individual year-to-year growth factors. This captures the idea that populations compound over time, rather than growing by the same fixed amount each year.

EDA of Instantaneous Rates

Countries with highest instantaneous rate over time.



Countries with lowest instantaneous rate over time.



Which countries have been experiencing the highest/lowest instantaneous rate in the last five years?

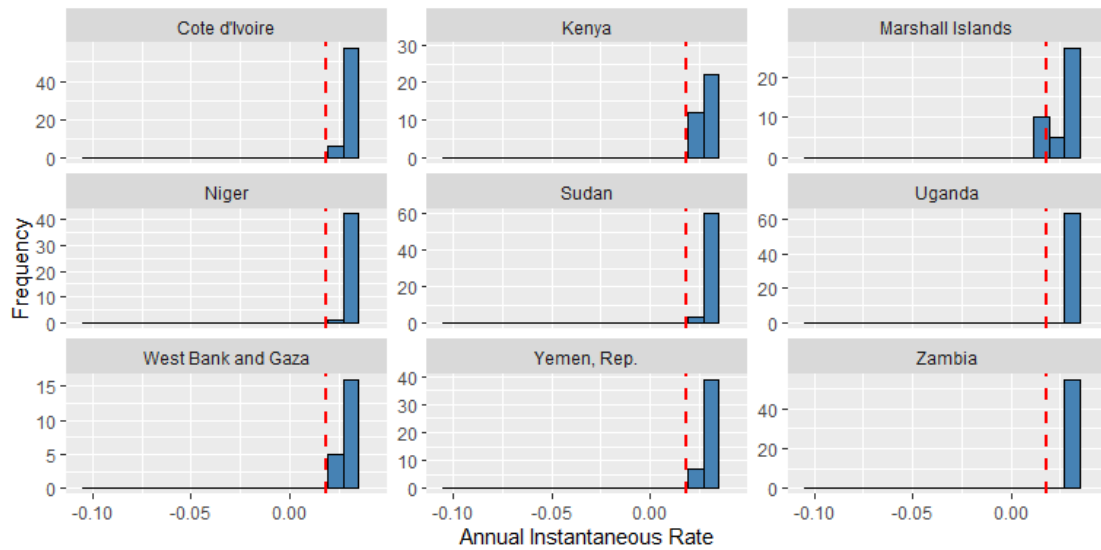
Top Countries highest increase in population in last 5 years

Name	Pop_2020	Pop_2024	mean_inst_rates
Niger	23,717,613	26,159,867	0.03281650
Congo, Dem. Rep.	95,989,998	105,789,731	0.03272750
Somalia, Fed. Rep.	16,651,191	18,358,615	0.03265250
Mali	21,713,836	23,769,127	0.03144375
Chad	17,224,679	19,319,064	0.03144300

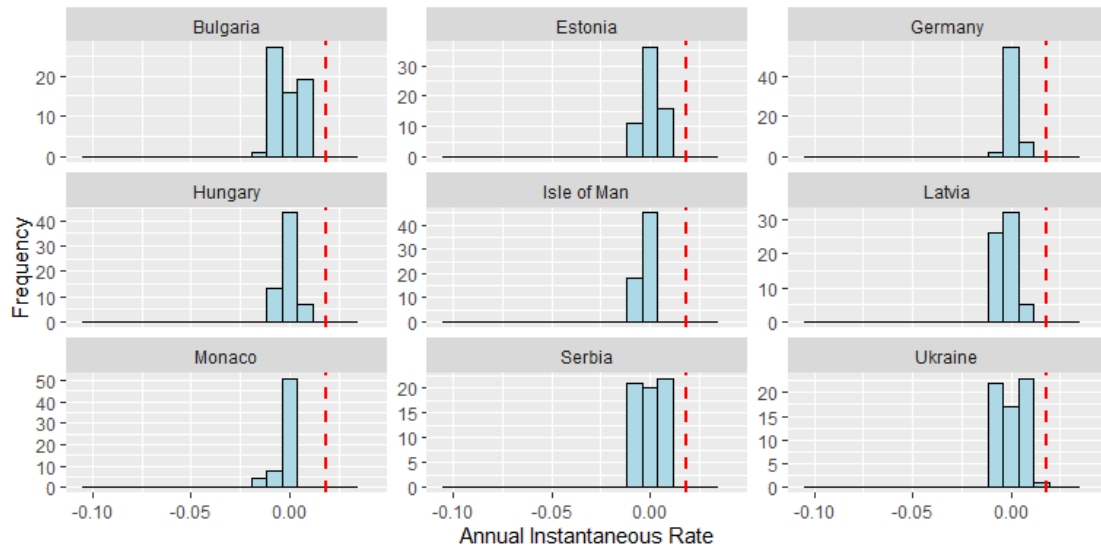
Countries with highest decrease in population in last 5 years

Name	Pop_2020	Pop_2024	mean_inst_rates
Monaco	38,050	38,956	-0.01343250
Bulgaria	6,550,696	6,446,596	-0.01007500
Ukraine	44,680,014	37,732,836	-0.00826350
Serbia	6,899,126	6,623,183	-0.00785000
Latvia	1,900,449	1,877,445	-0.00757500

Histograms of instantaneous rates with highest average rates over time.

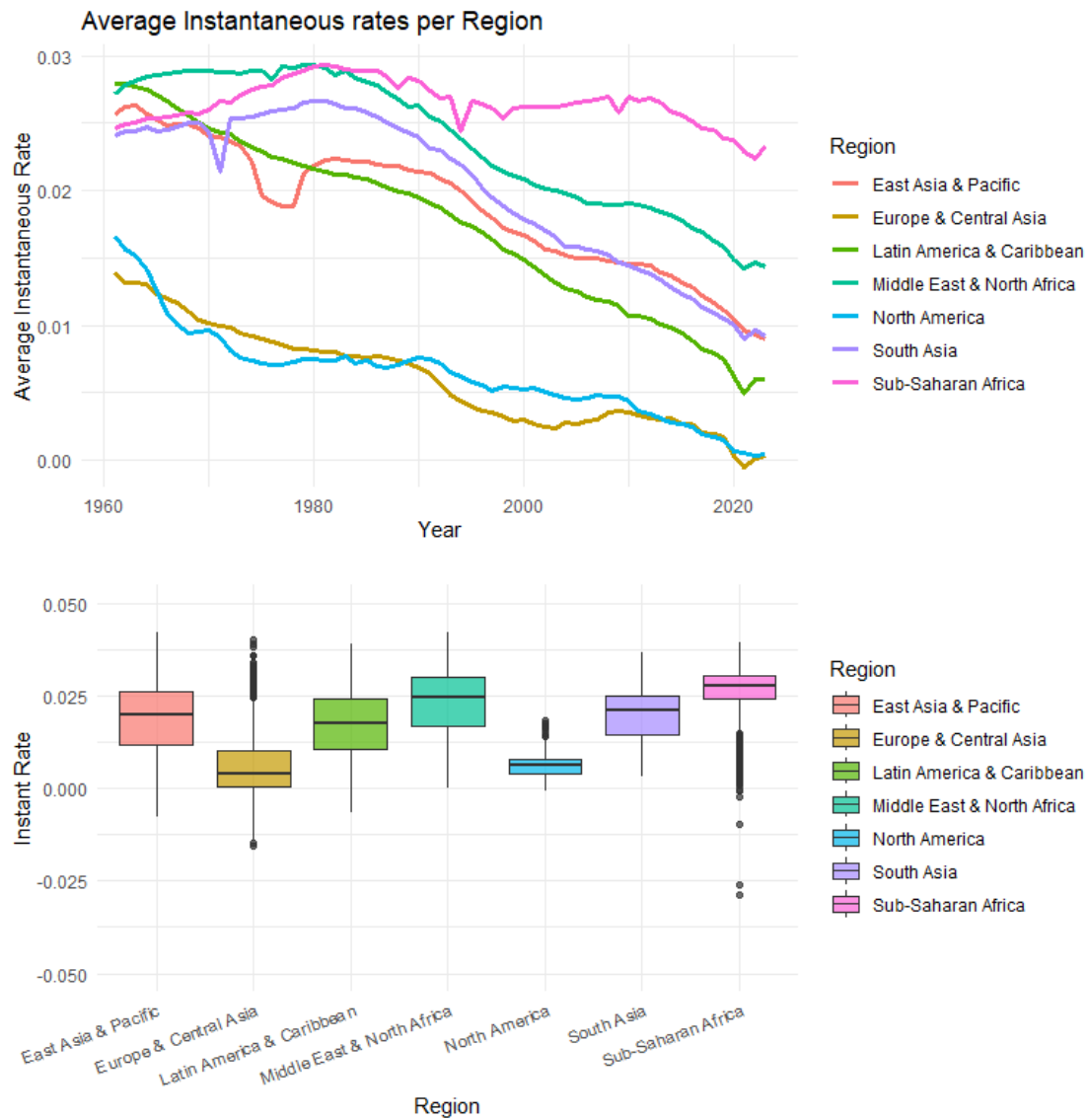


Histograms of instantaneous rates with lowest average rates over time.



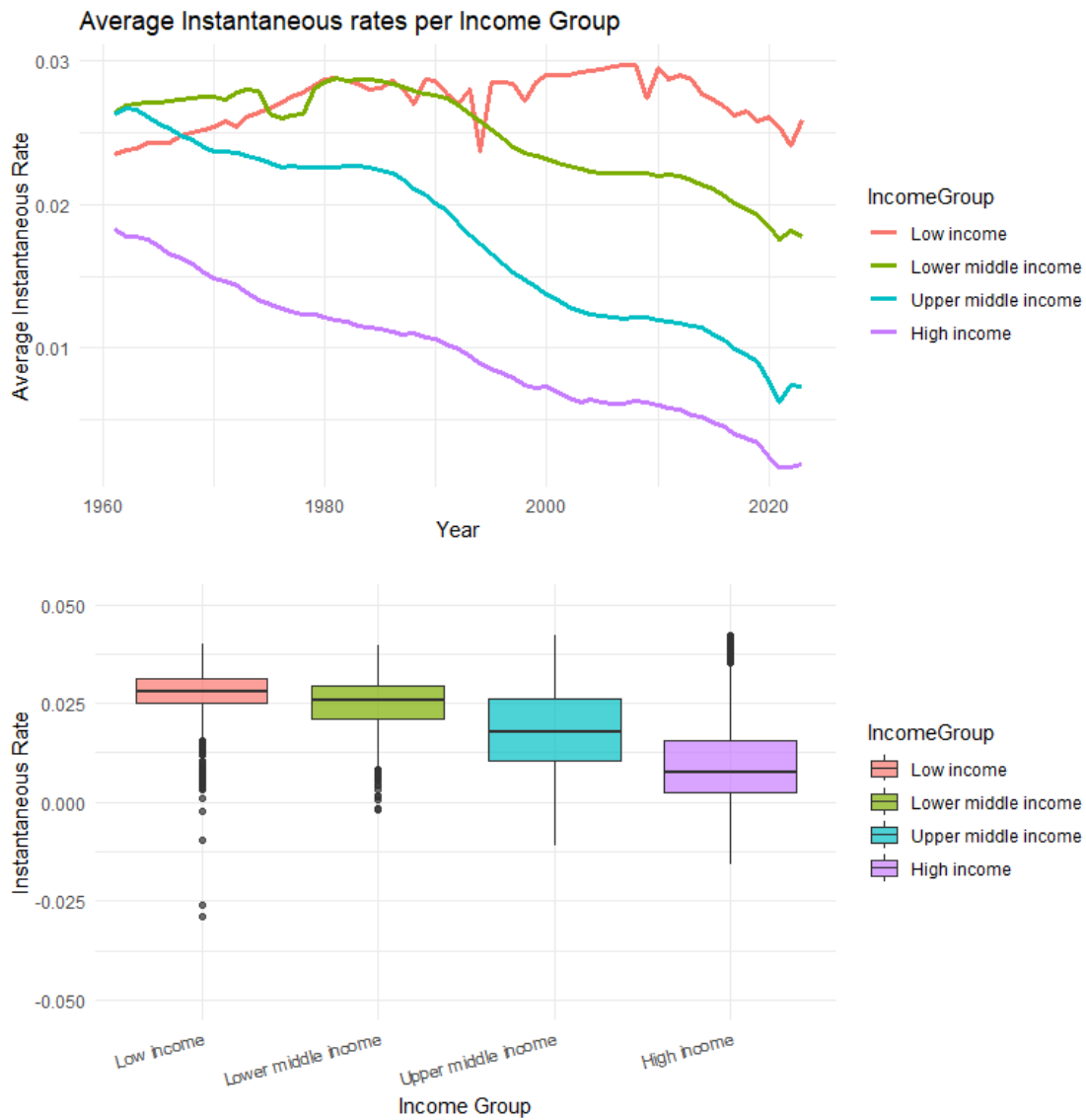
Countries with the highest instantaneous growth rates, such as Côte d'Ivoire, Kenya, Niger, Sudan, Uganda, Yemen, and Zambia, show distributions concentrated on positive values, indicating consistently strong natural population growth each year. In contrast, countries with the lowest instantaneous rates, including Bulgaria, Estonia, Hungary, Latvia, Serbia, and Ukraine, have distributions shifted toward zero or slightly negative values, reflecting slow growth or long-term population decline.

Instantaneous rates of countries over time grouped by region.



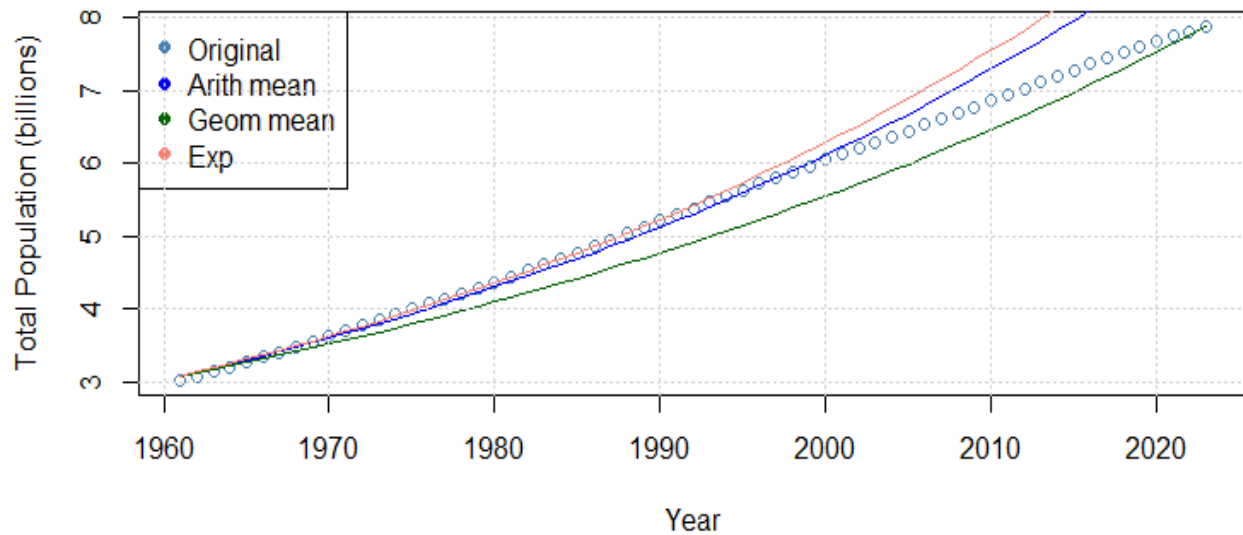
Across regions, average instantaneous growth rates have steadily declined since the 1960s, with the sharpest drops occurring in Europe & Central Asia and North America. Sub-Saharan Africa consistently shows the highest rates, followed by the Middle East, South Asia, and Latin America. The boxplots reinforce these patterns: Sub-Saharan Africa has the highest typical instantaneous rates, while Europe & Central Asia and North America have the lowest, reflecting clear regional differences in natural population growth.

Instantaneous rates of countries by income group.

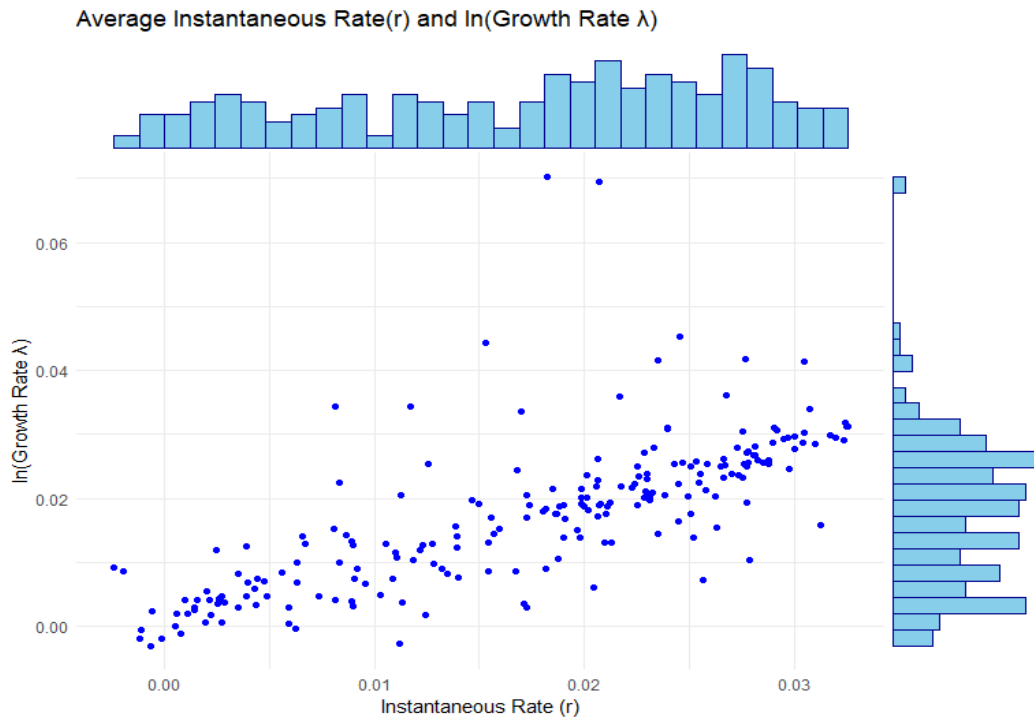


Instantaneous growth rates have declined over time across all income groups, but low-income and lower-middle-income countries consistently show the highest rates. Upper-middle-income countries have moderate rates that have fallen steadily since the 1980s, while high-income countries show the lowest and most stable rates. The boxplots highlight these differences clearly: growth is strongest in low- and lower-middle-income groups, weaker in upper-middle-income countries, and lowest in high-income countries.

Comparing the results in Parts 1 & 2



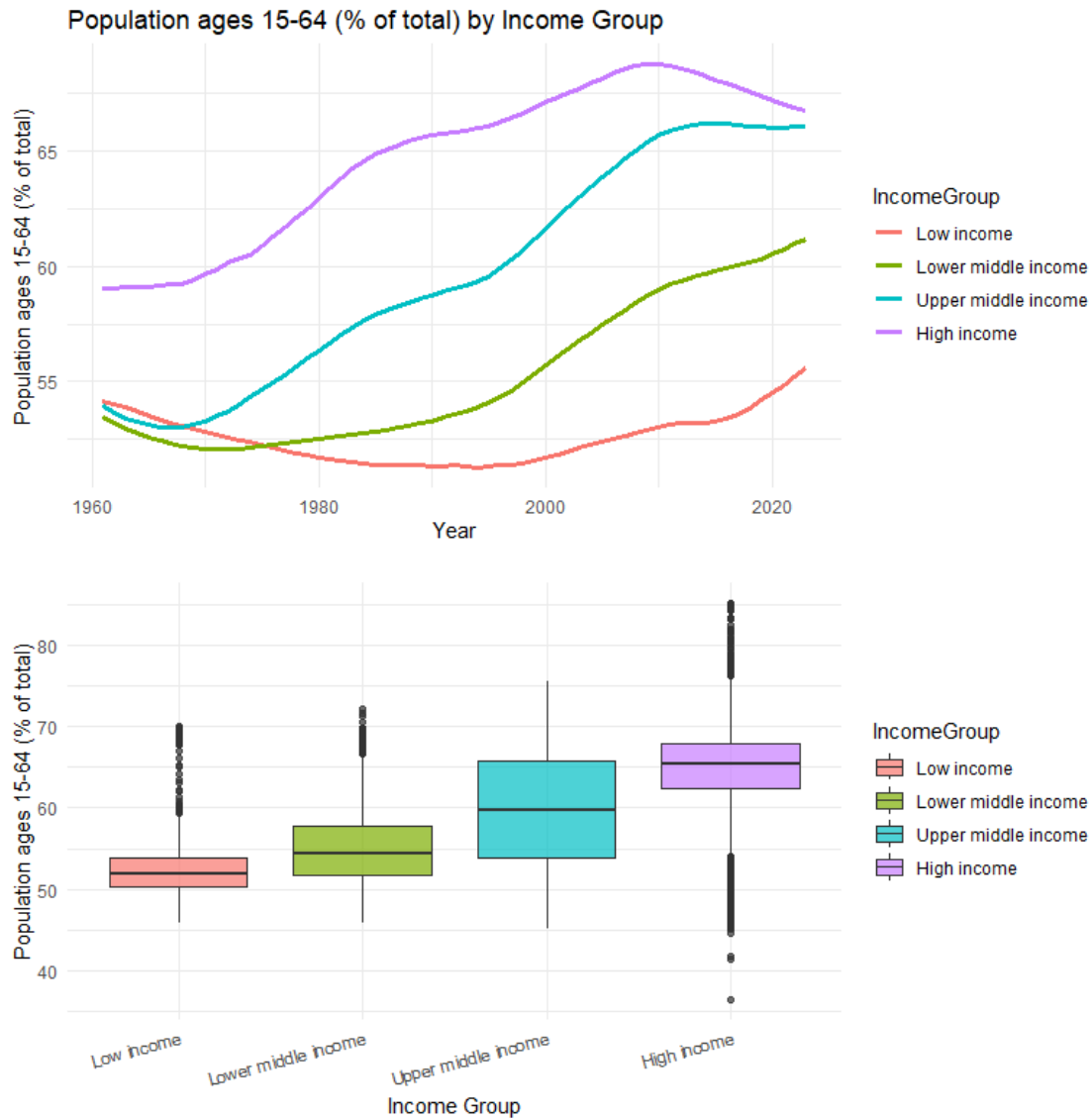
This plot compares the actual world population with three model-based projections: using the arithmetic mean growth rate, the geometric mean growth rate, and the exponential model. The arithmetic mean model grows the fastest and overestimates the population in later years, while the geometric mean model grows more slowly and underestimates the later values. The exponential model tracks the early trend well but also diverges upward over time.



The scatterplot shows a clear positive relationship between the average instantaneous rate and the logarithm of the annual growth rate. As instantaneous growth increases, the log of the growth factor also rises, which matches the theoretical link between the two measures.

Other Factors influencing growth: Migration and Age Distribution

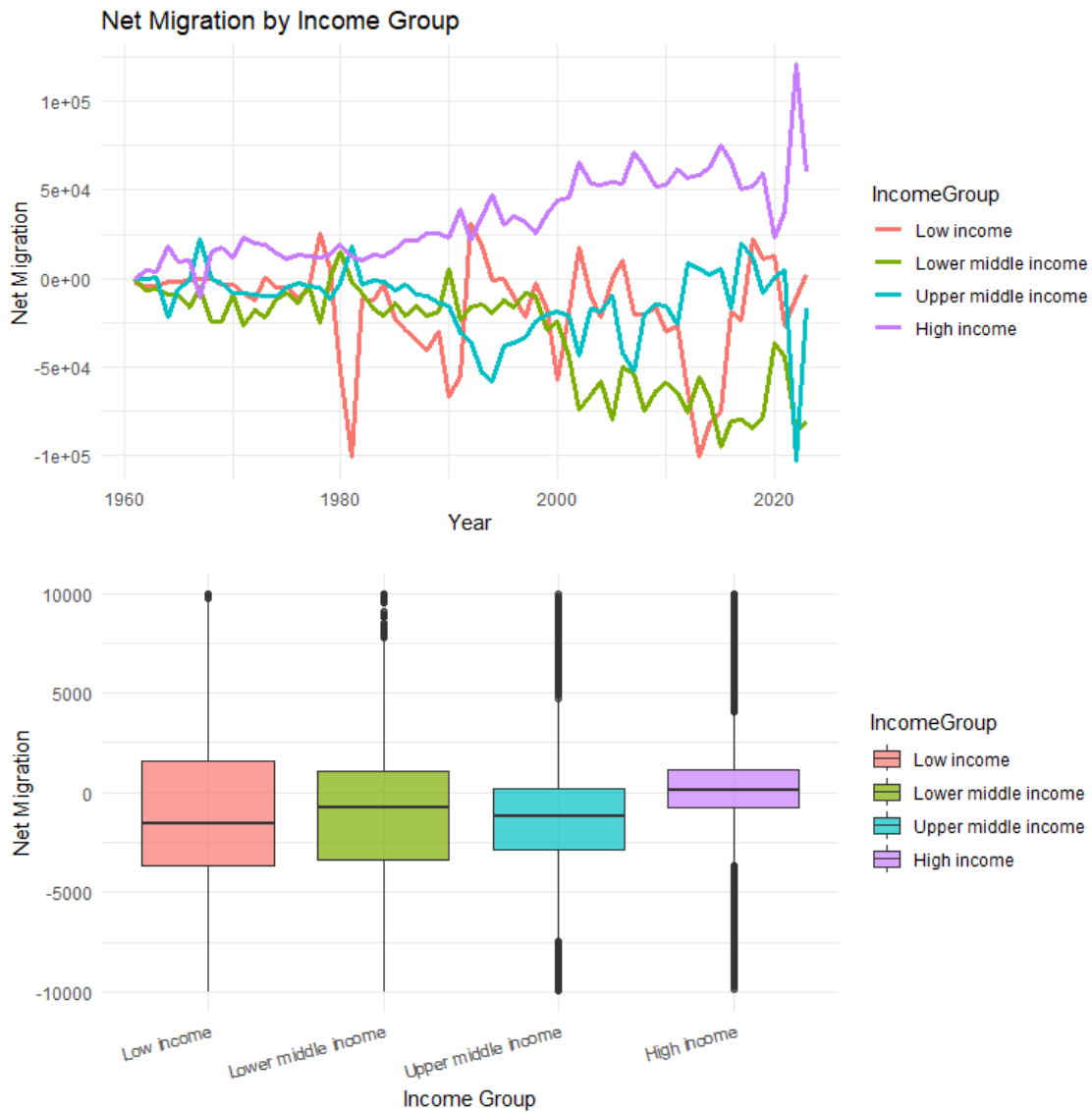
Age Distribution factor



Across income groups, the share of the population aged 15-64 has gradually risen over time, especially in upper-middle-income and high-income countries, which consistently maintain the highest working-age proportions. Lower-income countries show slower increases and remain below other groups. The boxplots reflect these differences clearly: high-income and upper-middle-income countries have the largest working-age shares, while low-income countries have the smallest, highlighting how demographic structure varies with economic development.

Migration Factor

Beyond natural increase, population change is also shaped by the movement of people across borders. Immigration adds to a population as people enter a country, while emigration reduces it as people leave. Net migration measures the balance between these two flows. A positive value means more people are arriving than departing, while a negative value means the opposite.

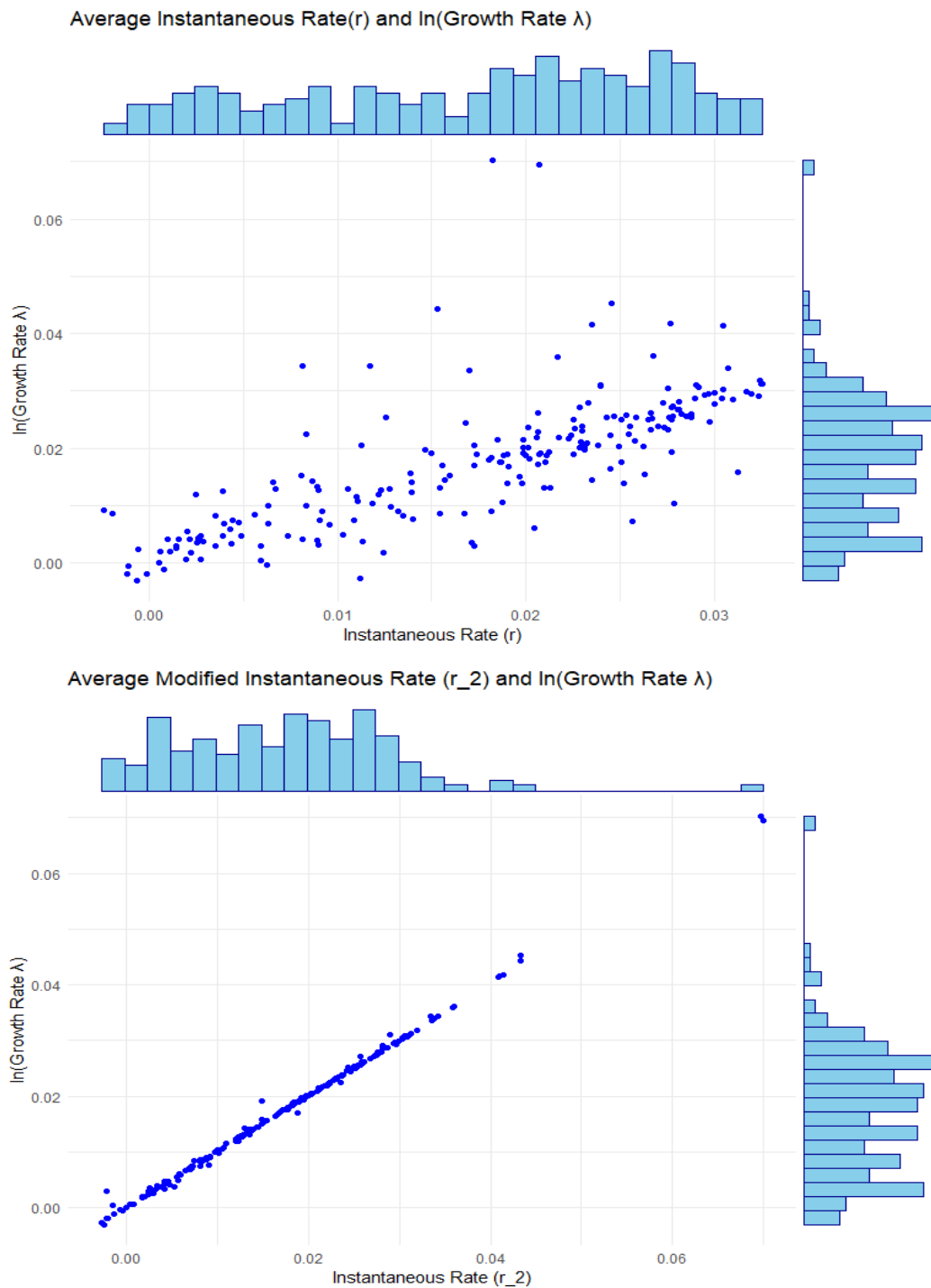


Looking at the raw net migration values, high-income countries consistently show the largest positive numbers, indicating steady inflows of migrants over time. In contrast, low-income, lower-middle-income, and upper-middle-income groups display much more fluctuation, often switching between positive and negative values. The boxplots reinforce this pattern: high-income countries typically receive more migrants than they lose, while other income groups tend to experience smaller or more variable net migration levels.

To compare migration levels across countries of different sizes, the net migration rate expresses this balance per 1,000 people in the population. It is calculated by taking the difference between immigrants and emigrants during a given period and dividing it by the mid-year population, then multiplying by 1,000.

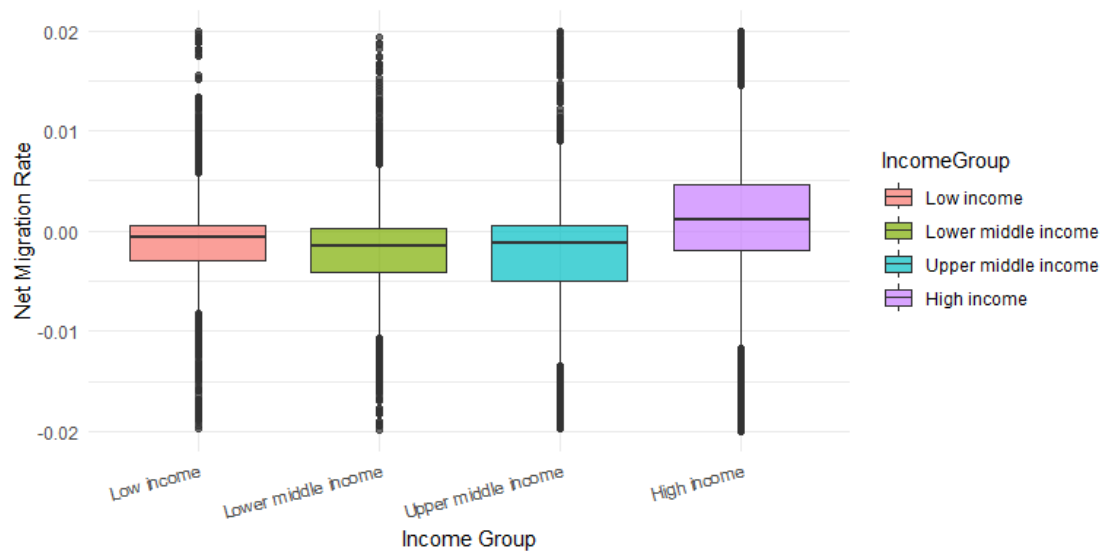


Net migration rates differ across income groups: high-income countries consistently show positive rates, meaning they attract more migrants, while low- and middle-income groups fluctuate around zero with frequent gains and losses. The boxplots confirm this pattern, with high-income countries having higher and more stable migration rates than the other groups.



Both plots show the relationship between the instantaneous growth rate and the log of the annual growth factor. The first plot shows a positive but somewhat scattered relationship, meaning the two measures align but with some variation across countries. After modifying the instantaneous rate, the second plot shows an almost perfect linear relationship, confirming the theoretical link between the two growth measures and demonstrating that the adjusted rate captures this connection much more precisely.

Hypothesis Testing



The boxplot comparing net migration across income groups indicates that high-income countries generally experience larger net inflows than the other groups. To confirm whether this visual pattern is statistically meaningful, we can apply a formal hypothesis testing approach.

```
##
## Welch Two Sample t-test
##
## data: High_Inc$mean_migration_rate and Up_Mid_Inc$mean_migration_rate
## t = 23.067, df = 91.999, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.00716948      Inf
## sample estimates:
##  mean of x    mean of y
##  0.003971228 -0.003754767
```

The test shows that High-Income countries have clearly higher migration rates than Upper-Middle-Income countries. The extremely small p-value indicates this difference is statistically significant, meaning High-Income countries generally gain more people through migration, while Upper-Middle-Income countries tend to lose people.

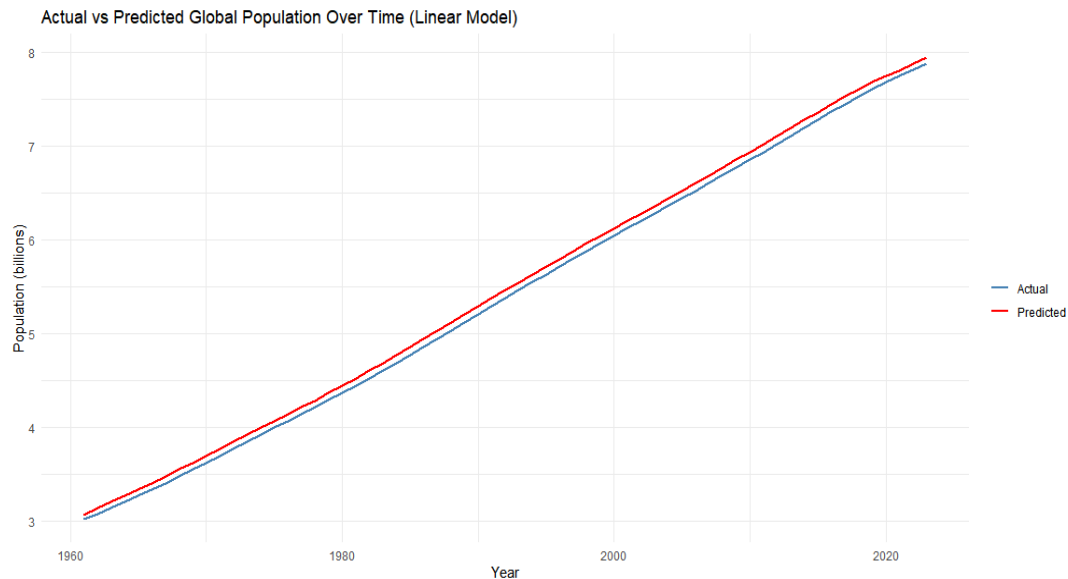
Part 3: Modeling

Linear Model for Instantaneous Rate

To understand which factors help explain variation in instantaneous population growth across countries, we fit a linear regression model using the computed instantaneous rate as the response variable. The predictors include birth rate, death rate, net migration rate, income group, and region. By incorporating both demographic variables and categorical groupings, this model allows us to evaluate how each contributes to differences in growth patterns and to quantify their combined effects on the instantaneous rate.

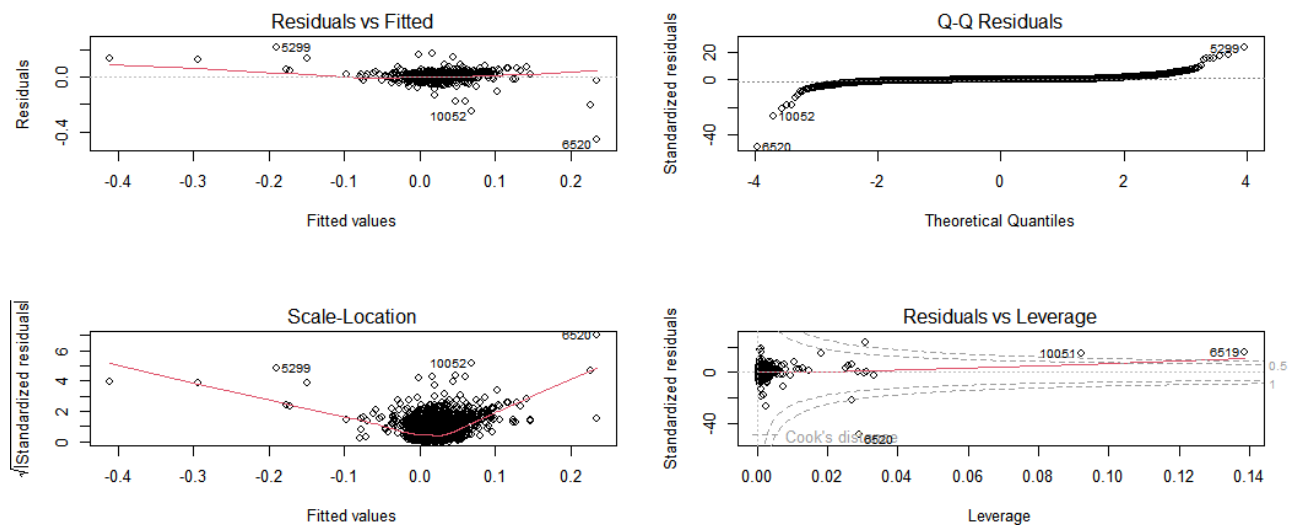
```
## Call:
## lm(formula = R_inst ~ BirthRate + DeathRate + Migration_Rate +
##     IncomeGroup + Region, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.45452 -0.00185 -0.00001  0.00159  0.21563
##
## Coefficients:
##                                Estimate Std. Error
## (Intercept)                   8.701e-04  3.124e-04
## BirthRate                     9.960e-04  1.074e-05
## DeathRate                     -9.675e-04  1.897e-05
## Migration_Rate                 6.252e-01  4.831e-03
## IncomeGroupLow income         -3.447e-03  4.001e-04
## IncomeGroupLower middle income -2.764e-03  2.902e-04
## IncomeGroupUpper middle income -2.695e-03  2.281e-04
## RegionEurope & Central Asia     2.070e-04  2.832e-04
## RegionLatin America & Caribbean -1.027e-04  2.746e-04
## RegionMiddle East, North Africa, Afghanistan & Pakistan 3.845e-03  3.265e-04
## RegionNorth America            1.960e-04  7.276e-04
## RegionSouth Asia               1.374e-03  5.399e-04
## RegionSub-Saharan Africa       1.286e-03  3.169e-04
##                                t value Pr(>|t|)
## (Intercept)                  2.785  0.00536 **
## BirthRate                   92.699 < 2e-16 ***
## DeathRate                   -50.989 < 2e-16 ***
## Migration_Rate             129.422 < 2e-16 ***
## IncomeGroupLow income       -8.616 < 2e-16 ***
## IncomeGroupLower middle income -9.523 < 2e-16 ***
## IncomeGroupUpper middle income -11.819 < 2e-16 ***
## RegionEurope & Central Asia    0.731  0.46490
## RegionLatin America & Caribbean -0.374  0.70846
## RegionMiddle East, North Africa, Afghanistan & Pakistan 11.777 < 2e-16 ***
## RegionNorth America          0.269  0.78759
## RegionSouth Asia             2.545  0.01094 *
## RegionSub-Saharan Africa      4.060  4.94e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.009428 on 13376 degrees of freedom
## (1 observation deleted due to missingness)
## Multiple R-squared:  0.7307, Adjusted R-squared:  0.7305
## F-statistic: 3025 on 12 and 13376 DF, p-value: < 2.2e-16
```

The model shows that population growth mainly depends on birth rates, death rates, and migration. Growth increases when birth rates or migration are higher and decreases when death rates are higher. Lower-income countries tend to grow more slowly than high-income countries even after accounting for these factors. Some regions, including the Middle East and North Africa, South Asia, and Sub-Saharan Africa, grow faster than the reference region, while Europe and North America show no major differences. Overall, the model explains about 73 percent of the differences in growth across countries and over years.



The model's predicted global population closely tracks the actual global population over time. The predicted line is almost identical to the actual line, which indicates that the linear model captures the long-term growth pattern very well. However, the model slightly overestimates population in most years, but the overall trend, slope, and shape match the real data closely. This suggests that the demographic variables used in the model provide strong predictive power for aggregate global population growth.

Diagnostic Plots for Linear Model



Overall, the diagnostic plots indicate that the linear model provides a reasonable fit. Residuals are centered and show no major structural pattern, suggesting the linearity assumption is broadly satisfied. The Q-Q plot reveals some tail deviations, and the Scale-Location plot shows mild heteroscedasticity, but neither issue is severe. No highly influential points appear in the leverage plot. These diagnostics suggest that, despite minor departures from ideal assumptions, the model is adequate for the purposes of this analysis.

Extended Linear Model for Instantaneous Rate

The extended model includes interactions between demographic variables and both income group and region. This allows the effects of birth rate, death rate, and migration to vary across economic and geographic categories, giving a more detailed view of how growth patterns differ between groups.

```
Call:
lm(formula = R_inst ~ (BirthRate + DeathRate + Migration_Rate) *
    (IncomeGroup + Region), data = data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.43896 -0.00162  0.00012  0.00143  0.17499
```

```
Coefficients:
(Intercept)                3.677e-03  6.650e-04   5.529  3.28e-08 ***
BirthRate                 9.647e-04  2.403e-05  40.140  < 2e-16 ***
DeathRate                -1.269e-03  7.288e-05 -17.410  < 2e-16 ***
Migration_Rate            4.399e-01  9.931e-03  44.298  < 2e-16 ***
IncomeGroupLow income     -3.966e-03  1.493e-03  -2.656  0.007909 **
IncomeGroupLower middle income -5.836e-03  9.411e-04  -6.201  5.77e-10 ***
IncomeGroupUpper middle income -3.931e-03  7.198e-04  -5.461  4.83e-08 ***
RegionEurope & Central Asia -1.013e-03  9.078e-04  -1.116  0.264444
RegionLatin America & Caribbean -1.088e-03  7.858e-04  -1.385  0.166089
RegionMiddle East, North Africa, Afghanistan & Pakistan 2.653e-03  9.021e-04   2.941  0.003274 **
RegionNorth America      -3.849e-03  6.686e-03  -0.576  0.564893
RegionSouth Asia         1.888e-03  1.659e-03   1.138  0.255001
RegionSub-Saharan Africa  3.003e-05  1.097e-03   0.027  0.978172
BirthRate:IncomeGroupLow income -1.800e-04  4.567e-05  -3.941  8.17e-05 ***
BirthRate:IncomeGroupLower middle income -9.719e-06  3.302e-05  -0.294  0.768487
BirthRate:IncomeGroupUpper middle income -7.428e-05  2.216e-05  -3.352  0.000805 ***
BirthRate:RegionEurope & Central Asia 6.514e-05  3.013e-05   2.162  0.030660 *
BirthRate:RegionLatin America & Caribbean 1.426e-04  2.948e-05  4.836  1.34e-06 ***
BirthRate:RegionMiddle East, North Africa, Afghanistan & Pakistan 2.248e-04  3.525e-05  6.378  1.85e-10 ***
BirthRate:RegionNorth America -4.703e-05  1.829e-04  -0.257  0.797065
BirthRate:RegionSouth Asia 5.900e-06  8.725e-05   0.068  0.946090
BirthRate:RegionSub-Saharan Africa 1.057e-04  3.638e-05   2.905  0.003680 **
DeathRate:IncomeGroupLow income 7.325e-04  8.326e-05  8.797  < 2e-16 ***
DeathRate:IncomeGroupLower middle income 4.398e-04  7.327e-05  6.002  2.01e-09 ***
DeathRate:IncomeGroupUpper middle income 5.011e-04  6.615e-05  7.575  3.83e-14 ***
DeathRate:RegionEurope & Central Asia 3.897e-05  8.577e-05   0.454  0.649565
DeathRate:RegionLatin America & Caribbean -2.369e-04  8.713e-05  -2.719  0.006549 **
DeathRate:RegionMiddle East, North Africa, Afghanistan & Pakistan -6.606e-04  7.539e-05  -8.763  < 2e-16 ***
DeathRate:RegionNorth America 4.433e-04  8.142e-04   0.544  0.586164
DeathRate:RegionSouth Asia -5.719e-05  1.620e-04  -0.353  0.724119
DeathRate:RegionSub-Saharan Africa -2.211e-04  6.415e-05  -3.447  0.000569 ***
Migration_Rate:IncomeGroupLow income 1.367e-02  2.336e-02  0.585  0.558408
Migration_Rate:IncomeGroupLower middle income 1.688e-01  1.698e-02  9.943  < 2e-16 ***
Migration_Rate:IncomeGroupUpper middle income 2.017e-01  1.780e-02  11.331  < 2e-16 ***
Migration_Rate:RegionEurope & Central Asia 2.743e-01  1.939e-02  14.142  < 2e-16 ***
Migration_Rate:RegionLatin America & Caribbean 4.087e-01  1.521e-02  26.866  < 2e-16 ***
Migration_Rate:RegionMiddle East, North Africa, Afghanistan & Pakistan 1.315e-01  1.217e-02  10.806  < 2e-16 ***
Migration_Rate:RegionNorth America 5.520e-01  1.770e-01  3.119  0.001816 **
Migration_Rate:RegionSouth Asia 1.648e-01  5.566e-02  2.961  0.003075 **
Migration_Rate:RegionSub-Saharan Africa 7.335e-02  2.435e-02  3.012  0.002597 **
```

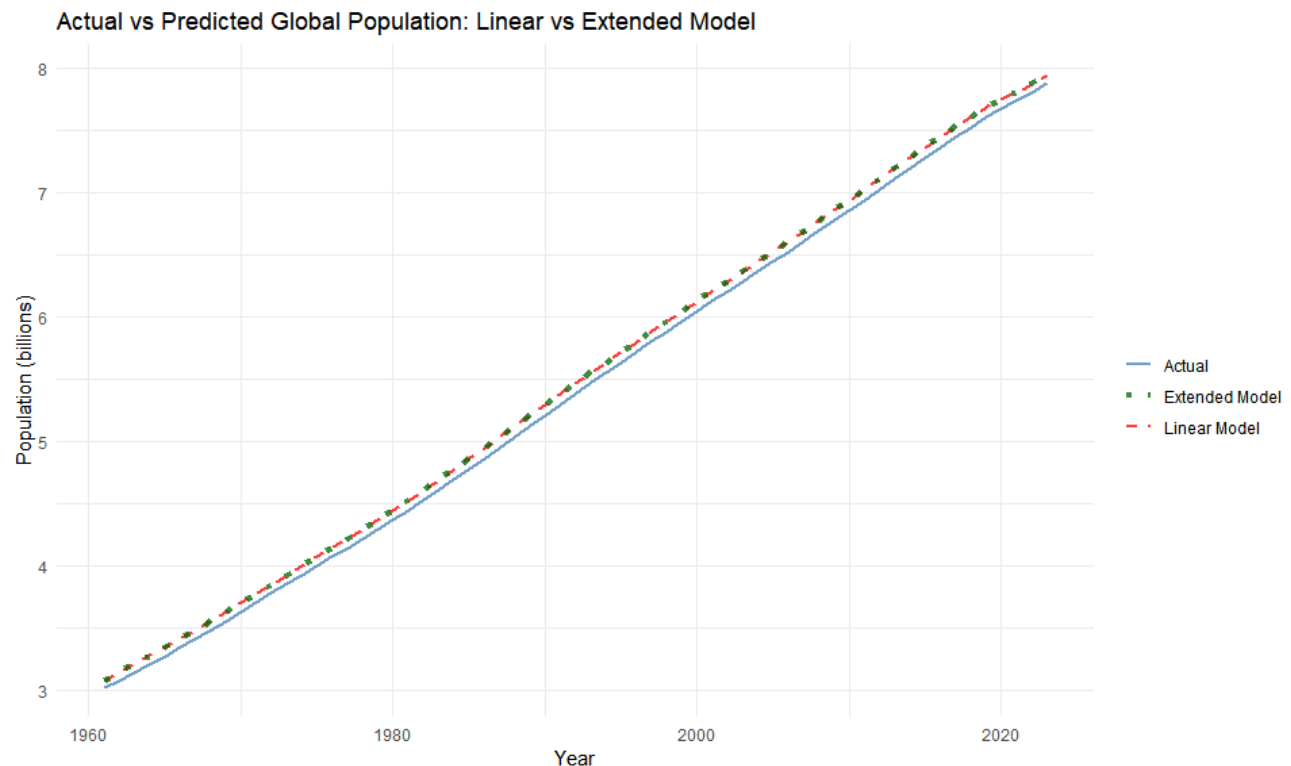
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.009001 on 13349 degrees of freedom
(1 observation deleted due to missingness)
Multiple R-squared:  0.755,    Adjusted R-squared:  0.7543
F-statistic: 1055 on 39 and 13349 DF,  p-value: < 2.2e-16
```

The extended model improves fit compared to the simpler linear model. The R-squared increases from about 0.73 to 0.755, indicating that allowing birth rates, death rates, and migration to interact with Income Group and Region explains additional variation in instantaneous growth rates.

Many interaction terms are statistically significant, which means the effect of demographic variables varies across income levels and regions.

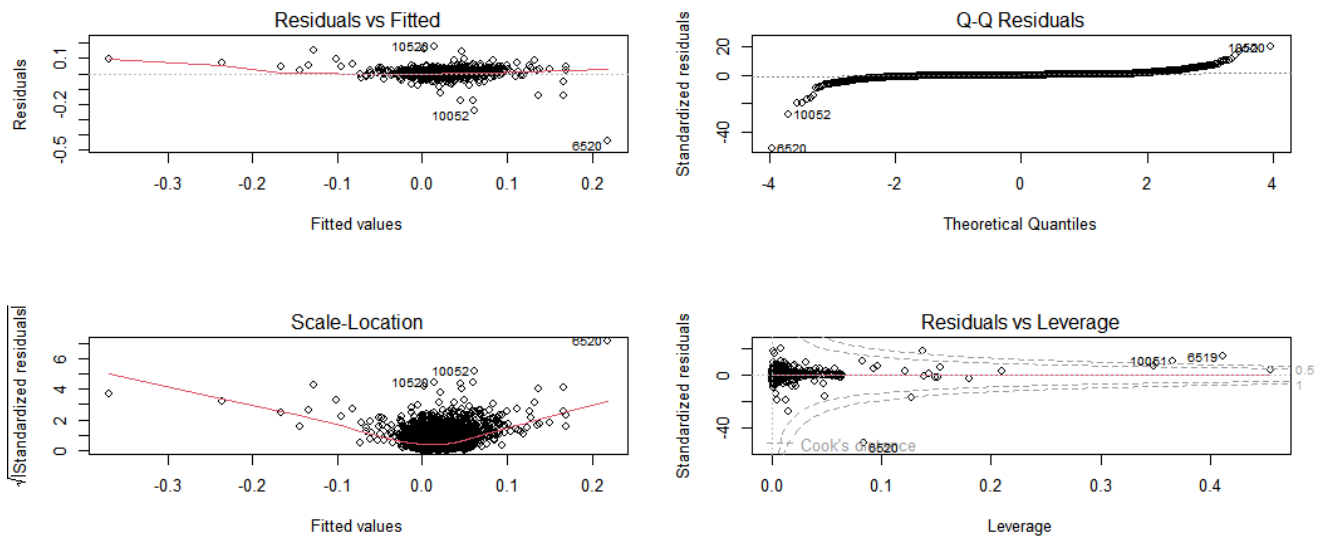
Overall, the extended model captures more nuanced demographic relationships and provides a better explanation of cross-country growth patterns.



The plot compares actual global population over time with the predictions from both the simple linear model and the extended model. Both models closely track the true population trend, indicating that the predictors explain global growth reasonably well.

However, the differences between the two models are relatively small, suggesting that even the simpler model captures the main pattern of long-term global population growth.

Diagnostic Plots for Extended Linear Model



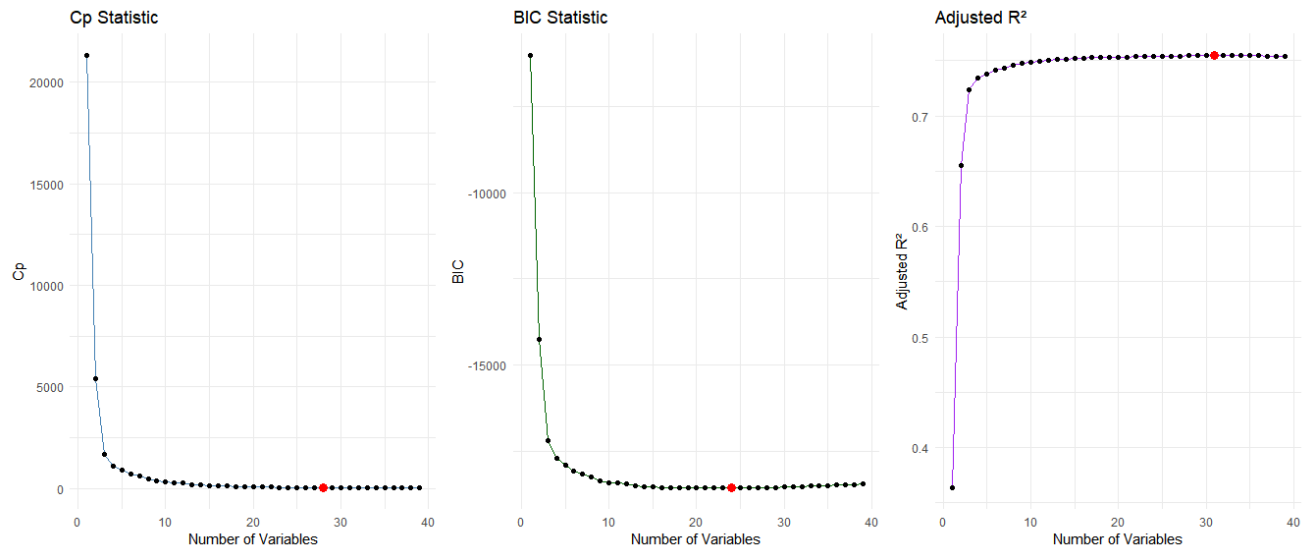
The extended linear model diagnostics show modest improvement but still reveal clear assumption violations. Residuals vs Fitted indicates remaining nonlinearity and uneven variance, and the Q-Q plot shows that residuals deviate from normality, especially in the tails. The Scale-Location plot confirms heteroscedasticity, while the Leverage plot highlights a few influential observations.

Comparing the two sets of diagnostics, the extended model reduces overall residual spread, showing that the interaction terms help capture more variation across regions and income groups.

However, one important difference is that the extended model produces noticeably higher-leverage observations, indicating that the added complexity increases sensitivity to certain countries or regions with extreme demographic patterns. This effect is much weaker in the simple linear model. In other words, the extended model fits the data better but relies more heavily on a few influential observations, which is a trade-off not visible when examining the extended model alone.

Subset Selection from The Extended Linear Model

To identify the most informative combination of predictors from the extended model, subset selection was performed using the *regsubsets* function. Model quality was evaluated using standard selection criteria, including Mallows' Cp, the Bayesian Information Criterion (BIC), and Adjusted R². These metrics help determine which subset of variables achieves the best balance between model fit and complexity.



The Cp and BIC curves drop sharply at first and then flatten, showing that most improvement comes from the early predictors. Adjusted R^2 increases quickly before leveling off, indicating limited benefit from adding more variables. Using BIC as the selection criterion, the model with 24 variables provides the best balance between fit and simplicity.

Forward and Backward Stepwise Selection

Forward and backward stepwise selection were applied to identify a more efficient set of predictors from the extended model. These methods iteratively add or remove variables based on improvement in model fit, allowing us to compare different paths toward a parsimonious model.

Backward Stepwise Selection

```
## Start:  AIC=-126094.4
## R_inst ~ (BirthRate + DeathRate + Migration_Rate) * (IncomeGroup +
##      Region)
##
##              Df Sum of Sq    RSS    AIC
## <none>                 1.0816 -126094
## - BirthRate:IncomeGroup    3  0.002033 1.0836 -126075
## - BirthRate:Region         6  0.004221 1.0858 -126054
## - DeathRate:Region         6  0.007665 1.0893 -126012
## - DeathRate:IncomeGroup    3  0.007509 1.0891 -126008
## - Migration_Rate:IncomeGroup 3  0.017136 1.0987 -125890
## - Migration_Rate:Region    6  0.064009 1.1456 -125337
```

Removing any of the interaction terms increases the AIC compared to the full model, meaning each term contributes to a better fit. Since no removal lowers AIC, the backward stepwise procedure keeps all interaction terms, confirming that the full extended model is already optimal under AIC.

Forward Stepwise Selection

Forward stepwise selection started with a model containing only the intercept and added predictors one at a time based on the largest improvement in AIC. The procedure first selected the main demographic variables, then added Region and IncomeGroup, and eventually incorporated the interaction terms that continued to improve the model. **Because the full sequence of stepwise outputs is extremely long, only the key steps are summarized here rather than listed in full.** The process ultimately stopped when no remaining variable reduced AIC, resulting in the same full extended model as the final selected model.

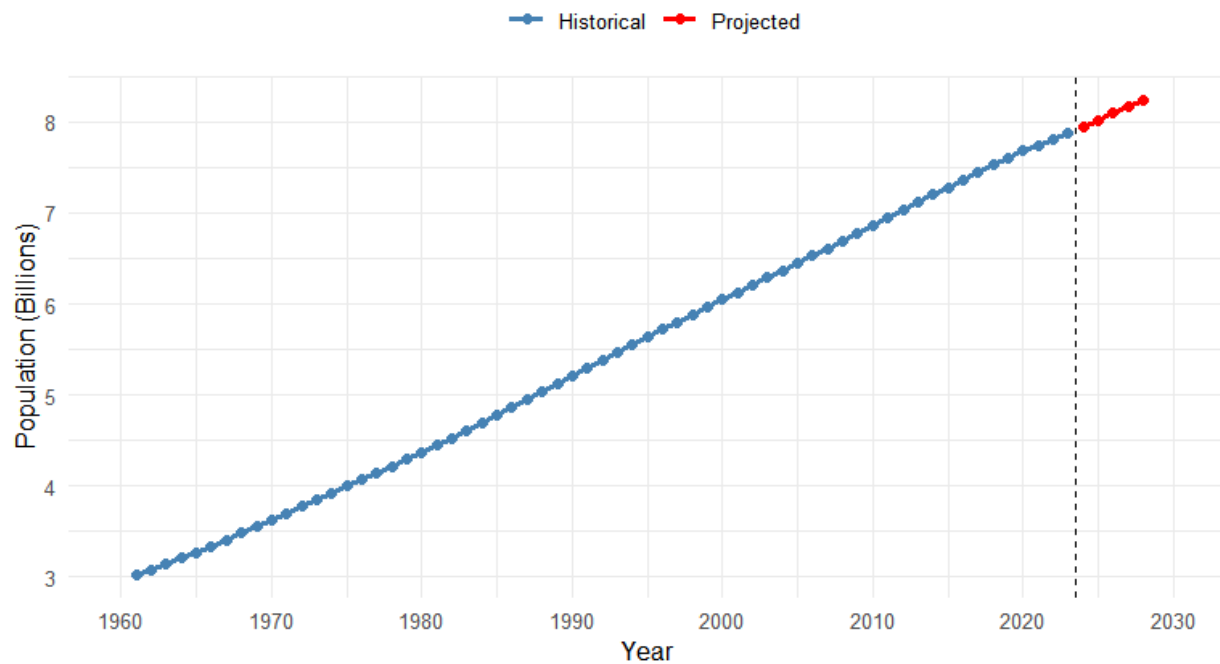
Model	AIC	BIC	Adj_R2
Extended Model	-88,096.08	-87,788.49	0.754
Backward Stepwise	-88,096.08	-87,788.49	0.754
Forward Stepwise	-88,096.08	-87,788.49	0.754

Projected Forecast

Base Year	Population Billion	Target Year	TY_Pop Billion	Difference Million	Growth_Pct
2023	7.883	2024	7.952	68.8	0.87
2023	7.883	2025	8.022	139.0	1.76
2023	7.883	2026	8.093	210.5	2.67
2023	7.883	2027	8.166	283.5	3.60
2023	7.883	2028	8.241	357.8	4.54

The forecast table shows projected global population from 2024 to 2028 using 2023 as the base year. World population is expected to rise from about 7.883 billion in 2023 to 8.241 billion by 2028. This represents an increase of roughly 358 million people over five years, corresponding to a total growth of about 4.5 percent. The year-to-year increments steadily increase, reflecting continued population expansion under the model's estimated growth rates.

World Population (1961-2028)



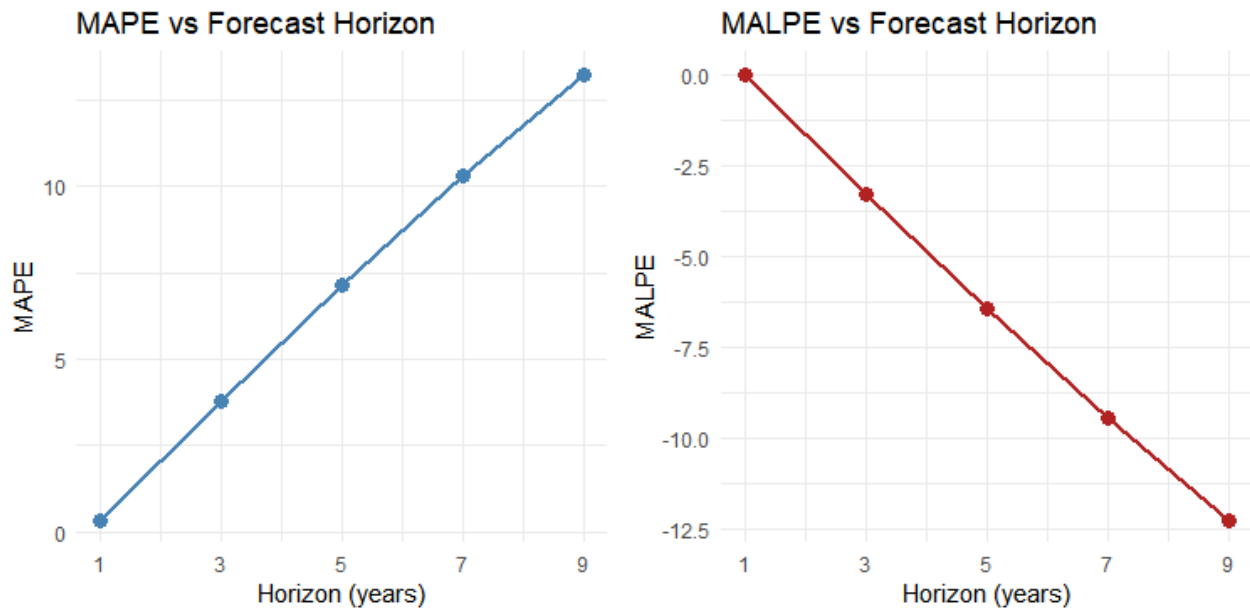
Part 4: Model Validation

Rolling Window Cross-Validation

To evaluate forecasting performance, rolling window cross-validation is applied. The model is repeatedly trained on data up to a selected base year and then used to predict population for a future target year. This window moves forward through time, creating multiple train-test splits. Forecast accuracy is assessed using several forecast horizons, in this case 1, 3, 5, 7, and 9 years ahead.

MAPE and MALPE are used to evaluate model performance. MAPE measures precision by reporting the average percent size of the forecast errors, ignoring direction. MALPE measures bias by keeping the sign of the errors to show whether forecasts tend to be too high or too low. Together, they give a clear picture of the model's accuracy and directional tendencies across different forecast lengths.

h	n	MAPE	MALPE
1	13,177	0.3286999	0.02448159
3	12,751	3.7672620	-3.30062870
5	12,325	7.1270854	-6.44394113
7	11,899	10.2787355	-9.43048462
9	11,473	13.2360065	-12.26977441

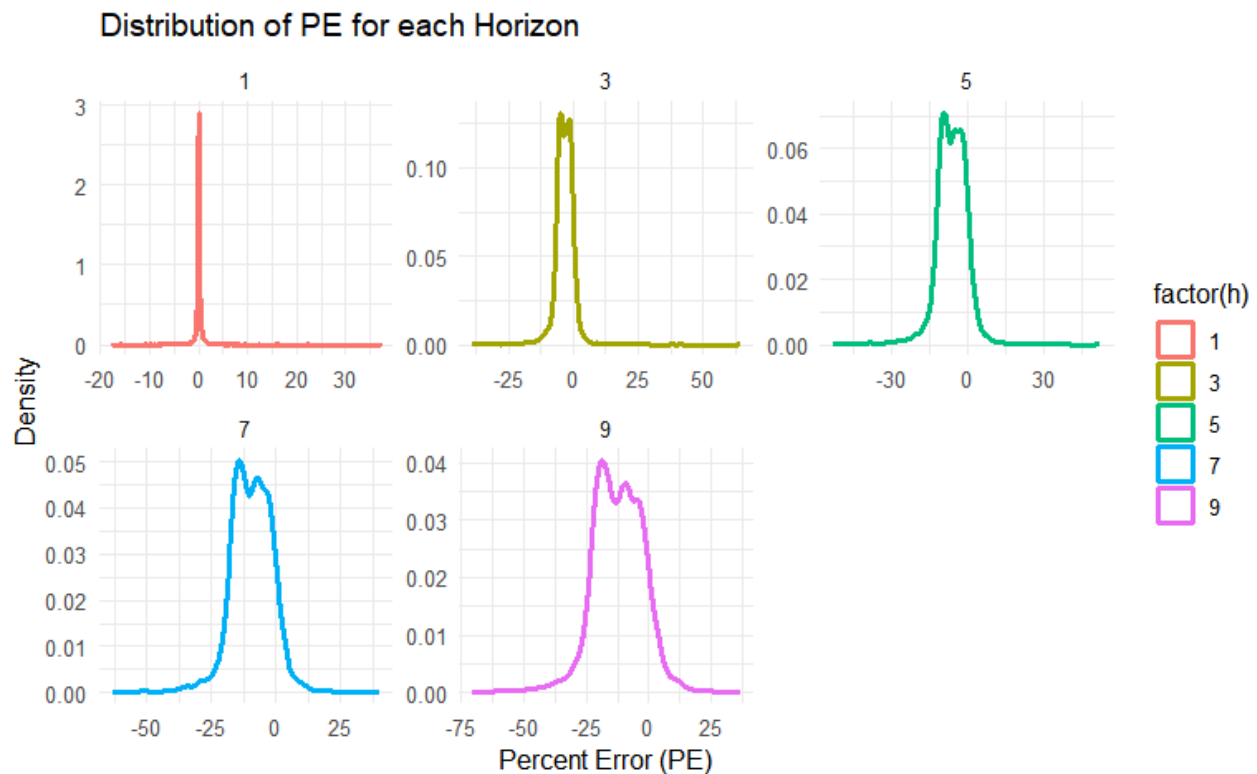


Forecast accuracy decreases as the horizon increases. For short-term forecasts ($h = 1$), the model performs very well, with a MAPE of about 0.33 percent and almost no bias. As the horizon extends to 3, 5, 7, and 9 years, MAPE rises steadily, reaching about 13 percent at the 9-year horizon. The MALPE values become increasingly negative, indicating that the model tends to underestimate population more heavily at longer horizons. Overall, the results show good short-term accuracy but growing error and downward bias as forecasts extend further into the future.

Different Behaviors in Training Fit and Forecasting Performance

Although the extended linear model fits the historical data very well, this strong **in-sample fit** does not guarantee equally strong **out-of-sample forecast performance**. The model diagnostics and fitted vs actual plots show close tracking of historical population trends, with slight overestimation in some periods. However, the rolling-window validation results reveal a different behavior: as the forecast horizon increases, the model increasingly **underestimates** future population, producing negative MALPE values.

This contrast arises because in-sample fit captures how well the model explains data it was trained on, whereas forecast bias reflects the model's ability to generalize beyond the observed period. A model can reproduce the historical curve accurately yet still fail to anticipate the continued upward trajectory of global population growth. This happens when growth patterns shift, when interactions behave differently outside the training window, or when the model's linear structure cannot fully capture long-run dynamics. Therefore, strong model fit does not preclude systematic forecast bias, and both must be evaluated to judge forecasting reliability.



The density plots show how forecast percent errors (PE) behave across different horizons. For $h = 1$, errors are extremely concentrated near zero, indicating very accurate short-term forecasts. At $h = 3$ and 5 , the distributions widen, showing increasing uncertainty, and the peaks shift slightly left, suggesting growing underestimation. By $h = 7$ and 9 , the error distributions become much wider and clearly left-skewed, indicating systematic underprediction and larger forecast variability. Overall, the longer the horizon, the greater the spread and the stronger the bias toward underestimating future population.

Limitations

- The linear models assume stable relationships between demographic variables over time, which may not hold during major social, economic, or political shifts.
- Migration data quality varies widely across countries, and inconsistent reporting can introduce bias into estimated growth rates.
- The extended model relies on many interaction terms, improving fit but increasing the risk of overfitting and model sensitivity to influential countries.
- Residual diagnostics reveal heteroscedasticity and non-normality, indicating that key linear regression assumptions are not fully satisfied.
- Forecast accuracy declines as the forecast horizon increases, reflecting challenges in projecting long-term demographic trends.
- Population forecasting does not account for unexpected events (e.g., pandemics, conflicts) that can significantly alter growth patterns.

Future Works

- Explore nonlinear or machine learning models (e.g., random forests, gradient boosting) to capture complex demographic interactions beyond linear assumptions.
- Incorporate additional predictors such as fertility rates, life expectancy, education levels, and urbanization to enhance explanatory power.
- Extending the forecasting approach to include probabilistic forecasts with confidence intervals rather than single-point estimates.
- Evaluate model robustness under scenario-based simulations (e.g., low-, medium-, high-growth UN variants).
- Perform more extensive cross-validation across longer historical periods to better understand long-term forecast stability.
- Investigate dynamic models that update parameters over time instead of relying on fixed-coefficient regression.

Wrap-Up

This project analyzed global population trends using World Bank data and showed consistent long-term growth across regions and income groups. Birth rates, death rates, and migration proved to be key drivers of population change, and the extended regression model captured important differences across countries. Short-term forecasts based on this model project steady increases in world population through 2028.

Validation through rolling-window cross-validation showed that the model predicts well for near-term horizons but becomes less accurate as the forecast window expands, tending to underestimate future population. This highlights the need for caution when extending demographic models too far beyond the data.

Overall, the study demonstrates that combining demographic indicators with statistical modeling provides a useful framework for understanding and forecasting global population growth.